

# NASA Climate Strategy & Space Observations

## NASA's Earth Science and Climate Strategy (2023-2026)

NASA plays a critical role in observing and understanding Earth's changing climate from space. By operating an extensive fleet of earth-observing satellites, the agency monitors key environmental indicators including global surface temperatures, greenhouse gas concentrations, ocean heat content, ice sheet mass, and sea-level rise.

Key satellite missions in NASA's climate portfolio include the Orbiting Carbon Observatory 2 (OCO-2) and OCO-3, which measure atmospheric carbon dioxide (CO<sub>2</sub>) with high precision and spatial resolution. These instruments allow scientists to track trends in net carbon emissions and identify localized CO<sub>2</sub> sources and sinks, providing valuable top-down verification for national greenhouse gas inventories.

Additionally, the Gravity Recovery and Climate Experiment Follow-On (GRACE-FO) mission tracks changes in Earth's gravity field to measure water movements and mass changes in ice sheets and glaciers. The Sentinel-6 Michael Freilich satellite provides continuous, highly accurate measurements of sea-level rise, continuing a decadal record that dates back to TOPEX/Poseidon in 1992.

Through these space-based observations, NASA supports international climate frameworks, including the UNFCCC's Global Stocktake, by delivering open-access data products that enable decision-makers to formulate effective mitigation and adaptation policies.